PROJECT REPORT

On

SECURE POST QUANTUM CLOUD STORAGE

For

"CSE 459 - CRYPTOGRAPHY AND NETWORK SECURITY"

Submitted by

S. Meghana – AP23110011415

O. Little Anji – AP23110011417

K. Akhila - AP23110011422

P. Arun Sai Kumar – AP23110011458

SEMESTER : VI
SECTION: Z

TO

Dr. Sanjay Kumar

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

of

SCHOOL OF ENGINEERING AND SCIENCES



SRM University-AP, Neerukonda, Andhra Pradesh 522240

April 2026

Certificate

Date: 28-Apr-26

This is to certify that the work present in this Project entitled **"SECURE POST-QUANTUM CLOUD STORAGE VAULT"** has been carried out by "**S. Meghana – AP23110011415 , O. Little Anji – AP23110011417** , **K. Akhila – AP2311001142 , P. Arun Sai Kumar – AP23110011458** " under my/our supervision. The work is genuine, original, and suitable for submission to the **SRM University – AP** for the award of **Bachelor of Technology in School of Engineering and Sciences**.

Supervisor

(Signature)

Dr Sanjay Kumar

Assistant Professor,

Department of Computer Science and Engineering,

SRM University-AP

ACKNOWLEDGEMENTS

We would like to express our sincere gratitude to everyone who has supported and guided us throughout the completion of our project titled **"Secure Post Quantum Cloud Storage."**

First and foremost, we are deeply thankful to our supervisor, **Dr. Sanjay Kumar, Assistant Professor, Department of Computer Science and Engineering, SRM University–AP**, for his invaluable guidance, expertise, and continuous encouragement. His insightful suggestions and support helped us understand the concepts of post-quantum cryptography and cloud security, and played a crucial role in the successful completion of this project.

We are also grateful to the Department of Computer Science and Engineering at SRM University – AP for providing the necessary resources, infrastructure, and a conducive environment that made this project possible. We sincerely thank all the faculty members for their guidance and support throughout our academic journey.

We extend our heartfelt appreciation to our team members for their cooperation, dedication, and collaborative efforts. Working together allowed us to effectively design and implement a secure cloud storage system using advanced cryptographic techniques.

Finally, we would like to thank our families and friends for their constant support, encouragement, and motivation, which inspired us to complete this project successfully.

This project is the result of collective effort, and we are truly thankful to everyone who contributed to its completion.

TABLE OF CONTENTS

CERTIFICATE.....
ACKNOWLEDGEMENTS.....
ABSTRACT.....

1. INTRODUCTION
- **1.1 BACKGROUND**.....
- **1.2 PROBLEM STATEMENT**.....
- **1.3 OBJECTIVES**.....
- **1.4 SCOPE OF THE PROJECT**.....

2. METHODOLOGY.....
- **2.1 SYSTEM ARCHITECTURE**.....
- **2.2 ALGORITHM AND PROCESS FLOW**.....
- **2.3 IMPLEMENTATION DETAILS**.....
- **2.4 SECURITY FEATURES**.....
- **2.5 SYSTEM WORKFLOW**.....
- **2.6 ADVANTAGES OF THE PROPOSED METHOD**.....

3. DISCUSSION.....
- **3.1 RESULTS**.....
- **3.2 COMPARATIVE ANALYSIS**.....
- **3.3 OBSERVATIONS AND CHALLENGES**.....
- **3.4 SECURITY ANALYSIS**.....
- **3.5 USER PERSPECTIVE**.....

4. CONCLUDING REMARKS.....
- **4.1 SUMMARY OF ACHIEVEMENTS**.....
- **4.2 KEY INSIGHTS**.....
- **4.3 PRACTICAL IMPLICATIONS**.....
- **4.4 CHALLENGES ENCOUNTERED**.....
- **4.5 FINAL THOUGHTS**.....

5. FUTURE WORK.....

REFERENCES.....

ABSTRACT

This project presents the design and implementation of a **Secure Post-Quantum Cloud Storage Vault**, developed to ensure the confidentiality, integrity, and security of data stored in cloud environments. With the rapid growth of cloud computing, protecting sensitive information from unauthorized access and future cyber threats has become a major concern.

The proposed system utilizes a hybrid cryptographic approach by combining both symmetric and asymmetric encryption techniques. The file data is encrypted using **AES-256-GCM**, which provides high-speed encryption along with authentication to prevent data tampering. To secure the encryption keys, the system supports two mechanisms: **RSA-2048**, a widely used classical cryptographic algorithm, and **ML-KEM-768**, a post-quantum key encapsulation mechanism designed to resist attacks from quantum computers.

To further enhance data integrity, the system generates a **SHA-256 hash** of the file before encryption and verifies it after decryption. This ensures that any modification or corruption of the data can be detected reliably. The entire system is implemented as a web-based application using Flask and SQLite, allowing users to securely upload, store, and retrieve files through an intuitive interface.

Additionally, performance benchmarking is conducted to compare the efficiency of RSA and ML-KEM approaches. The results indicate that the post-quantum ML-KEM method provides competitive performance while offering stronger resistance against future quantum threats.

Overall, this project demonstrates a practical and scalable solution for secure cloud storage by integrating modern cryptographic techniques with post-quantum security, making it suitable for next-generation data protection systems.

1. INTRODUCTION

Cloud computing has revolutionized the way data is stored, accessed, and managed by providing scalable and cost-effective storage solutions. With the rapid growth of cloud services, individuals and organizations increasingly rely on remote servers to store sensitive and critical data. However, this convenience also introduces significant security challenges, particularly concerning data confidentiality, integrity, and unauthorized access.

The growing dependence on cloud platforms has made data security one of the most critical aspects of modern computing. Sensitive information such as personal data, financial records, and organizational documents are frequently stored in cloud environments, making them attractive targets for cyberattacks. Ensuring that this data remains secure during storage and transmission is a major concern for both users and service providers.

1.1 BACKGROUND

Traditional cryptographic algorithms such as RSA and Elliptic Curve Cryptography (ECC) have long been used to secure data in cloud environments. These algorithms rely on mathematical problems that are computationally difficult for classical computers to solve. However, with the advancement of quantum computing, these cryptographic techniques are becoming vulnerable. Quantum algorithms, such as Shor's algorithm, can efficiently break widely used encryption methods, posing a serious threat to data security.

To address this challenge, Post-Quantum Cryptography (PQC) has emerged as a promising solution. PQC focuses on developing cryptographic algorithms that are resistant to attacks from both classical and quantum computers. These algorithms are based on hard mathematical problems such as lattice-based, hash-based, and code-based cryptography, which are believed to be secure even in the presence of quantum computing.

1.2 PROBLEM STATEMENT

The increasing dependency on cloud storage combined with the potential threat of quantum computing creates a critical need for enhanced security mechanisms. Existing encryption techniques may become obsolete in the near future, making sensitive data vulnerable to attacks. Therefore, there is a need to design a secure cloud storage system that can withstand both current and future cryptographic threats, ensuring long-term data protection.

1.3 OBJECTIVES

The main objectives of this project are:

- To design a secure cloud storage system using post-quantum cryptographic techniques
- To ensure data confidentiality by encrypting files before uploading them to the cloud
- To implement secure key generation, encryption, and decryption mechanisms
- To prevent unauthorized access to stored data
- To provide a future-proof solution against quantum-based attacks

1.4 SCOPE OF THE PROJECT

This project focuses on implementing a secure data storage system where files are encrypted using post-quantum cryptographic algorithms before being uploaded to the cloud. The system ensures that only authorized users with valid keys can access and decrypt the data. It can be extended to real-world cloud platforms and integrated with advanced security mechanisms for enhanced protection.

2. METHODOLOGY

The methodology of the proposed **Secure Post Quantum Cloud Storage system** focuses on designing and implementing a robust security framework that protects data against both classical and quantum computing threats. The system ensures that sensitive data is encrypted using post-quantum cryptographic techniques before being stored in the cloud, thereby maintaining confidentiality, integrity, and secure access.

The overall workflow includes user authentication, key generation, encryption, secure storage, retrieval, and decryption. Each phase is carefully designed to ensure maximum security and efficiency.

2.1 SYSTEM ARCHITECTURE

The system follows a modular architecture consisting of multiple interconnected components that work together to provide secure cloud storage.

1. User Interface Layer :

This layer acts as the interaction point between the user and the system. It allows users to:

- Register and log in securely
- Upload files for storage
- Download and access stored files

The interface ensures ease of use while maintaining security protocols.

2. Authentication Module :

This module verifies the identity of users before granting access to the system. It includes:

- Username and password verification
- Session management
- Access control mechanisms

Only authenticated users are allowed to perform operations such as upload and download.

3. Key Management Module :

This is a critical component responsible for generating and managing cryptographic keys. It includes:

- Public key generation for encryption
- Private key generation for decryption
- Secure storage and handling of keys

Post-quantum algorithms are used to ensure resistance against quantum attacks.

4. Encryption Module :

Before uploading data to the cloud, the system encrypts the file using the public key. This module:

- Converts plaintext data into ciphertext
- Uses secure post-quantum encryption algorithms
- Ensures that data is unreadable without proper authorization

5. Cloud Storage Module :

This module stores the encrypted data. Key features include:

- Storage of ciphertext instead of plaintext
- Secure file handling
- Prevention of unauthorized data access

Even if the storage is compromised, the data remains secure due to encryption.

6. Decryption Module :

When a user requests a file, this module:

- Retrieves encrypted data from storage
- Decrypts it using the private key
- Converts it back into readable format

2.2 ALGORITHM AND PROCESS FLOW

The system follows a step-by-step secure process:

Step 1: User Authentication

The user logs into the system using valid credentials. Authentication ensures that only authorized users can access the system.

Step 2: Key Generation

A pair of cryptographic keys (public and private) is generated using post-quantum cryptographic algorithms.

- Public key → used for encryption
- Private key → used for decryption

These keys are securely stored and managed.

Step 3: File Selection and Processing

The user selects a file for upload. The file is read and converted into a suitable format (binary data) for encryption.

Step 4: Data Encryption

The file is encrypted using the public key.

- Plaintext → Ciphertext
- Ensures confidentiality of data

The encryption process uses post-quantum secure algorithms to protect against future quantum threats.

Step 5: Secure Storage

The encrypted file is uploaded to the cloud storage system.

- Only ciphertext is stored
- No sensitive information is exposed

Step 6: File Retrieval

When the user requests the file:

- The encrypted file is fetched from storage
- Access is verified before retrieval

Step 7: Data Decryption

The encrypted file is decrypted using the private key:

- Ciphertext → Plaintext
- Original file is restored

Step 8: Output Display

The decrypted file is presented to the user in its original format.

2.3 IMPLEMENTATION DETAILS

The system is implemented using modern programming tools and cryptographic libraries.

Programming Environment :

- Python / Java / C++ (based on your project)
- Supports file handling and cryptographic operations

Technologies Used:

- Post-Quantum Cryptographic Algorithms
- File Processing Techniques
- Cloud Storage Simulation / APIs

Modules Implemented:

- User authentication module
- Encryption and decryption module
- File upload/download system
- Key management system

Data Handling:

- Files are processed as binary streams
- Keys are securely generated and stored
- Encryption ensures secure transformation of data

2.4 SECURITY FEATURES

The system incorporates multiple layers of security:

Confidentiality

Data is encrypted before storage, ensuring that unauthorized users cannot access it.

Integrity

Ensures that data remains unchanged during transmission and storage.

Authentication

Only verified users can access and manipulate data.

Quantum Resistance

The use of post-quantum algorithms ensures protection against future quantum-based attacks.

2.5 SYSTEM WORKFLOW

- User opens the application and logs in using valid credentials
- Authentication module verifies the user identity
- System generates or retrieves the user's public and private keys
- User selects a file for upload
- File is read and converted into a suitable format (binary data)
- Encryption module encrypts the file using the public key
- Plaintext is converted into ciphertext
- Encrypted file is uploaded to cloud storage

- Only encrypted data is stored in the cloud
- User requests to download or access the stored file
- System verifies user authorization
- Encrypted file is retrieved from cloud storage
- Decryption module decrypts the file using the private key
- Ciphertext is converted back into original plaintext
- Decrypted file is displayed or downloaded by the user

2.6 ADVANTAGES OF THE PROPOSED METHOD

- Provides strong security against quantum attacks
- Ensures secure cloud data storage
- Protects sensitive information from unauthorized access
- Future-proof cryptographic solution
- Scalable and adaptable to real-world applications

3. DISCUSSION

The proposed **Secure Post Quantum Cloud Storage system** was successfully designed and implemented to ensure secure data storage using post-quantum cryptographic techniques. The system was evaluated based on its ability to protect data confidentiality, integrity, and resistance to future quantum-based attacks.

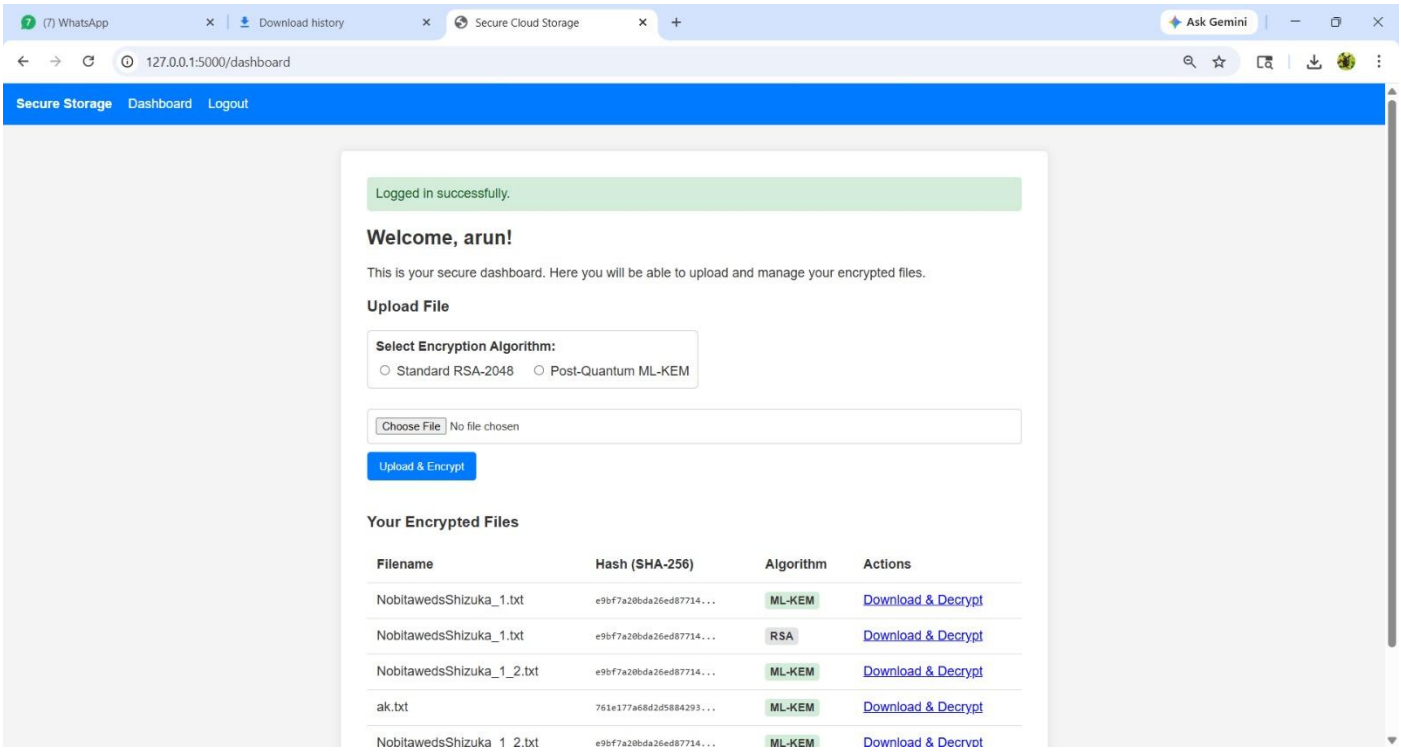
3.1 RESULTS

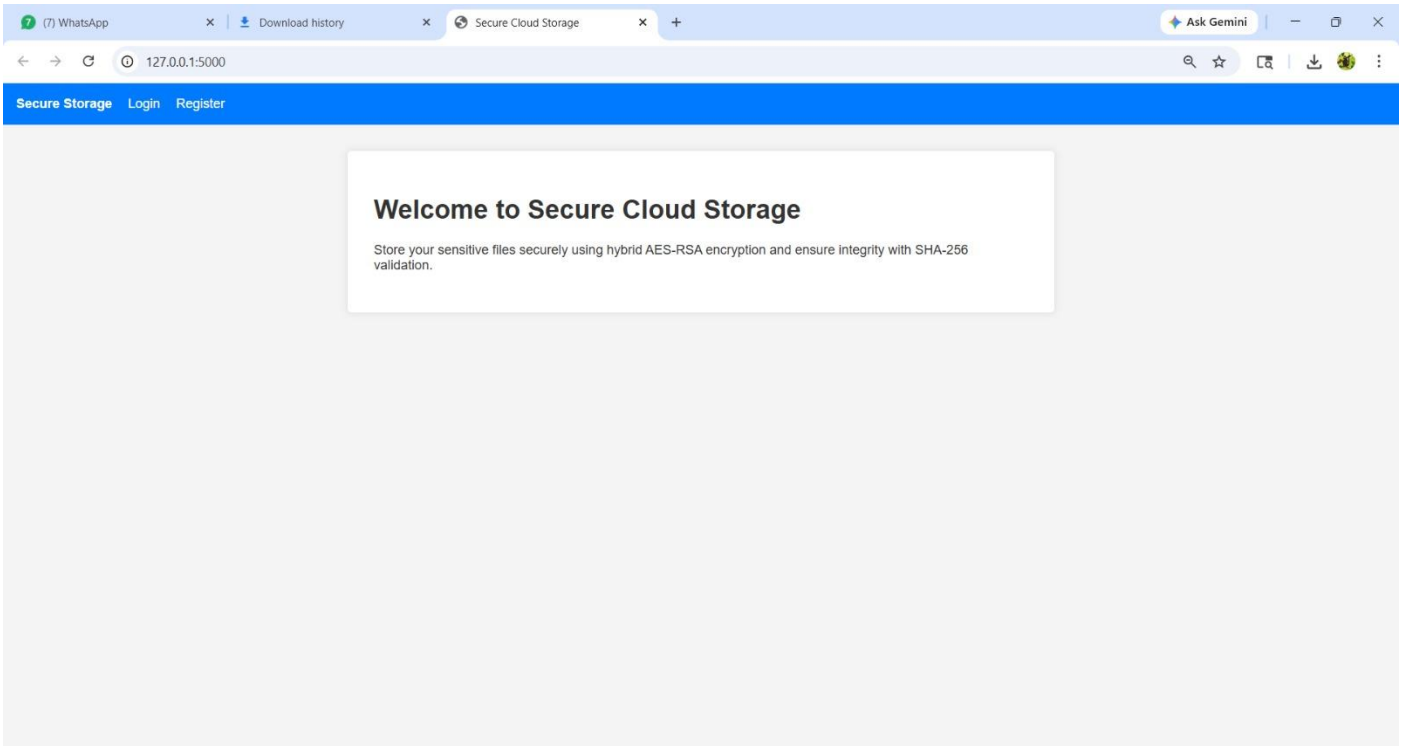
The system demonstrated effective performance in securely storing and retrieving data. The following outcomes were observed:

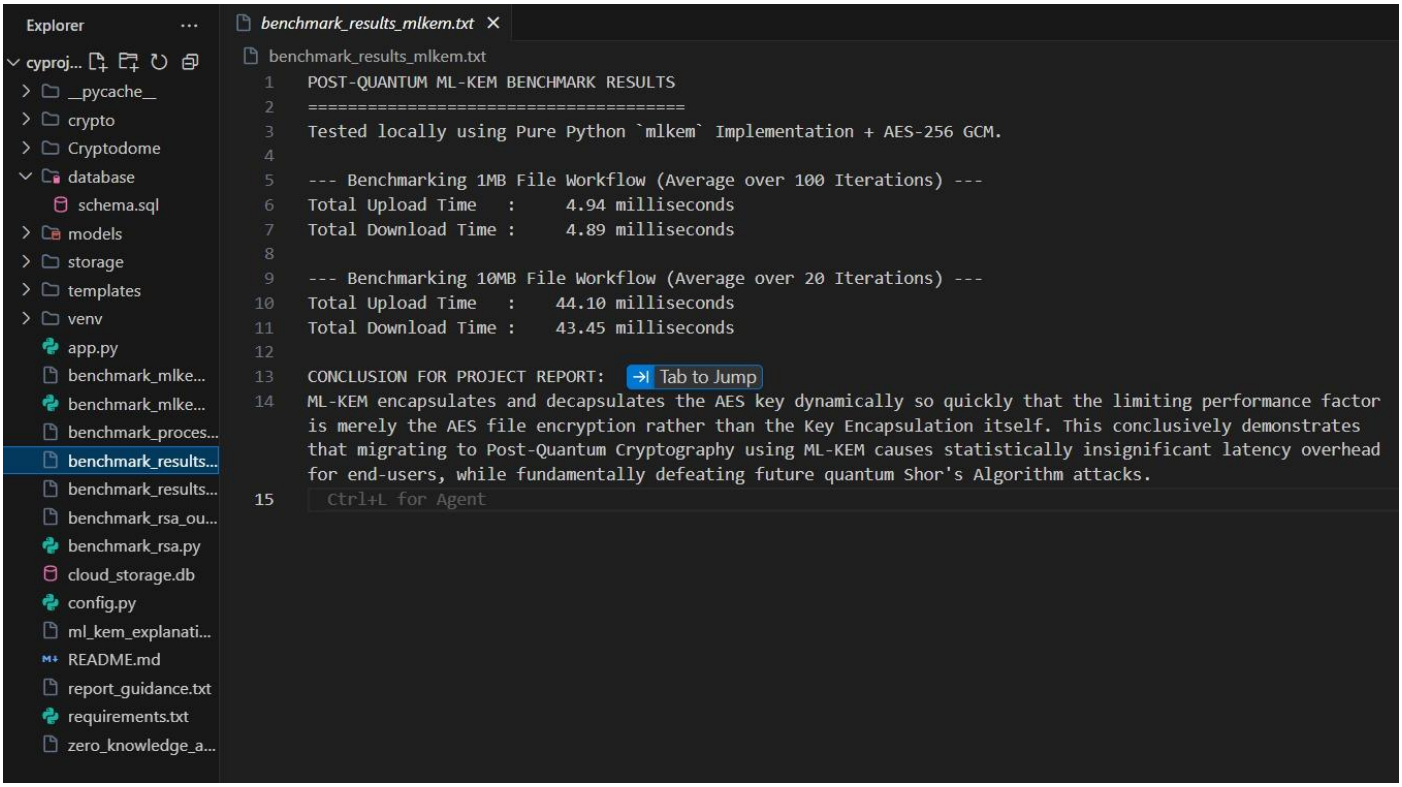
- Files were successfully encrypted before being uploaded to the cloud
- Only authorized users were able to decrypt and access the data
- Data remained secure even when stored in an untrusted environment
- The system maintained confidentiality by storing only encrypted data

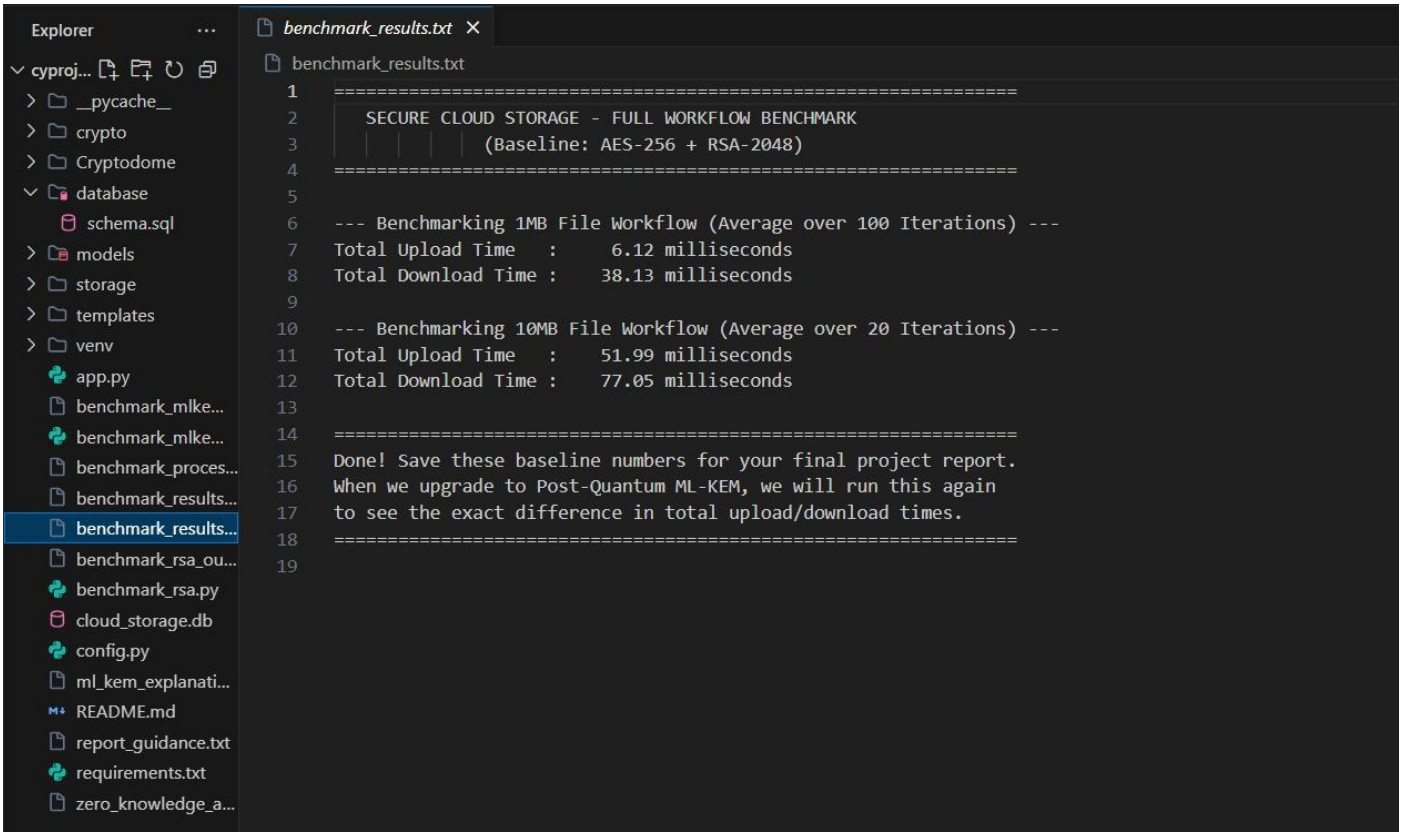
The implementation confirmed that post-quantum cryptographic techniques can be effectively applied in cloud storage systems.

Output:




(7) WhatsApp
Download history
Secure Cloud Storage
Ask Gemini
127.0.0.1:5000
Secure Storage
Login
Register
Welcome to Secure Cloud Storage
Store your sensitive files securely using hybrid AES-RSA encryption and ensure integrity with SHA-256 validation.


Explorer
cyproj...
__pycache__
crypto
Cryptodome
database
schema.sql
models
storage
templates
venv
app.py
benchmark_mlke...
benchmark_mlke...
benchmark_proces...
benchmark_results...
benchmark_results...
benchmark_rsa_ou...
benchmark_rsa.py
cloud_storage.db
config.py
ml_kem_explanati...
README.md
report_guidance.txt
requirements.txt
zero_knowledge_a...
benchmark_results_mlkem.txt
benchmark_results_mlkem.txt
POST-QUANTUM ML-KEM BENCHMARK RESULTS
Tested locally using Pure Python `mlkem` Implementation + AES-256 GCM.
--- Benchmarking 1MB File Workflow (Average over 100 Iterations) ---
Total Upload Time : 4.94 milliseconds
Total Download Time : 4.89 milliseconds
--- Benchmarking 10MB File Workflow (Average over 20 Iterations) ---
Total Upload Time : 44.10 milliseconds
Total Download Time : 43.45 milliseconds
CONCLUSION FOR PROJECT REPORT:
Tab to Jump
ML-KEM encapsulates and decapsulates the AES key dynamically so quickly that the limiting performance factor is merely the AES file encryption rather than the Key Encapsulation itself. This conclusively demonstrates that migrating to Post-Quantum Cryptography using ML-KEM causes statistically insignificant latency overhead for end-users, while fundamentally defeating future quantum Shor's Algorithm attacks.
Ctrl+L for Agent



3.2 COMPARATIVE ANALYSIS

A comparison between traditional cryptographic methods and post-quantum cryptography highlights the advantages of the proposed system:

- **Security:** Traditional algorithms like RSA and ECC are vulnerable to quantum attacks, whereas post-quantum algorithms provide strong resistance
- **Future Readiness:** The proposed system is designed to withstand future quantum threats
- **Performance:** Post-quantum algorithms may have slightly higher computational overhead compared to traditional methods
- **Key Size:** PQC algorithms generally use larger key sizes, increasing storage requirements

Overall, the proposed approach offers better long-term security despite minor performance trade-offs.

3.3 OBSERVATIONS AND CHALLENGES

During the development and testing of the system, several observations and challenges were identified:

- Post-quantum algorithms require more computational resources compared to traditional methods
- Key management is more complex due to larger key sizes
- Ensuring efficient encryption and decryption without performance degradation was challenging
- Handling large files required optimization to maintain system performance

Despite these challenges, the system successfully achieved its objective of secure cloud storage.

3.4 SECURITY ANALYSIS

The system incorporates multiple layers of security to ensure data protection:

- **Confidentiality:** Achieved through strong encryption techniques
- **Integrity:** Ensures that data is not altered during storage or transmission
- **Authentication:** Only verified users can access the system
- **Quantum Resistance:** Protects against attacks from future quantum computers

These features make the system robust and reliable for secure data storage.

3.5 USER PERSPECTIVE

From a user's point of view, the system provides:

- Simple and secure file upload and download process
- Strong protection of sensitive data
- Reliable access control mechanisms
- Confidence in long-term data security

The system balances usability and security, making it suitable for practical applications.

4. CONCLUDING REMARKS

The Secure Post Quantum Cloud Storage system developed in this project provides a robust and future-oriented solution for safeguarding data in cloud environments. With the rapid advancement of quantum computing, traditional cryptographic techniques are becoming increasingly vulnerable. This project addresses that concern by integrating post-quantum cryptographic methods to ensure long-term data security.

The system demonstrates how sensitive data can be securely encrypted before being stored in the cloud and accessed only by authorized users through proper decryption mechanisms. By combining cloud storage with advanced encryption techniques, the project ensures confidentiality, integrity, and controlled access to data.

4.1 SUMMARY OF ACHIEVEMENTS

The project successfully designed and implemented a secure framework for cloud storage where data is encrypted before being uploaded and decrypted only by authorized users. The system effectively ensures confidentiality, integrity, and controlled access to stored data.

4.2 KEY INSIGHTS

- Post-quantum cryptography provides stronger security compared to traditional methods
- Encrypting data before storage significantly reduces security risks
- Quantum computing poses a serious threat to existing cryptographic systems
- Advanced encryption techniques are essential for future data protection

4.3 PRACTICAL IMPLICATIONS

- Can be applied in real-world cloud storage systems
- Useful for securing sensitive data such as financial and personal information
- Can be extended to secure communication systems and enterprise applications
- Helps organizations prepare for future quantum computing threats

4.4 CHALLENGES ENCOUNTERED

- Higher computational complexity compared to traditional encryption methods
- Larger key sizes in post-quantum algorithms
- Efficient key management was challenging
- Performance optimization required for handling large data

4.5 FINAL THOUGHTS

- The project demonstrates the importance of quantum-resistant security solutions
- Provides a strong foundation for future secure cloud systems
- Encourages adoption of post-quantum cryptography in modern applications
- Highlights the need for continuous improvement in cybersecurity technologies

5. FUTURE WORK

The **Secure Post Quantum Cloud Storage system** developed in this project provides a strong foundation for secure and quantum-resistant data storage. However, several enhancements can be made to improve its efficiency, scalability, and usability.

1. **INTEGRATION WITH REAL CLOUD PLATFORMS**
   The current system can be extended by integrating it with real-world cloud service providers such as AWS, Microsoft Azure, or Google Cloud. This will enable large-scale deployment and practical usage in real-time environments.

2. **PERFORMANCE OPTIMIZATION**
   Post-quantum cryptographic algorithms often involve higher computational complexity. Future work can focus on optimizing these algorithms to improve encryption and decryption speed, especially for large files and real-time applications.

3. **HYBRID CRYPTOGRAPHIC APPROACH**
   A hybrid approach combining traditional cryptographic techniques with post-quantum algorithms can be implemented to achieve a balance between performance and security. This would enhance efficiency while maintaining strong protection against quantum attacks.

4. **GRAPHICAL USER INTERFACE (GUI)**
   Developing a user-friendly graphical interface will improve the usability of the system. A GUI can provide easy navigation for uploading, downloading, and managing files securely.

5. **ADVANCED SECURITY FEATURES**
   The system can be enhanced by incorporating additional security mechanisms such as:
   1. Multi-factor authentication
   2. Biometric verification
   3. Role-based access control

   These features will further strengthen user authentication and data protection.

6. **BLOCKCHAIN INTEGRATION**
   Integrating blockchain technology can improve transparency and security in key management and data access. It can provide tamper-proof logs and secure tracking of file transactions.

7. **SUPPORT FOR MULTIPLE PQC ALGORITHMS**
   Future work can include implementing and comparing different post-quantum cryptographic algorithms to evaluate their performance, efficiency, and security levels.

REFERENCES

1. National Institute of Standards and Technology (NIST),
   Post-Quantum Cryptography Standardization Project,
   Available: https://csrc.nist.gov/projects/post-quantum-cryptography

2. W. Stallings,
   Cryptography and Network Security: Principles and Practice,
   Pearson Education, 7th Edition.

3. D. J. Bernstein, J. Buchmann, and E. Dahmen,
   Post-Quantum Cryptography,
   Springer, 2009.

4. GeeksforGeeks,
   Introduction to Cryptography and Network Security,
   Available: https://www.geeksforgeeks.org/cryptography-and-network-security/

5. GeeksforGeeks,
   Cloud Computing Security Issues,
   Available: https://www.geeksforgeeks.org/cloud-computing-security/

6. IBM Research,
   Quantum Computing and Cryptography,
   Available: https://www.ibm.com/quantum

7. Microsoft Azure Documentation,
   Cloud Security and Encryption,
   Available: https://learn.microsoft.com/en-us/azure/security/

8. Google Cloud Documentation,
   Data Encryption and Security Overview,
   Available: https://cloud.google.com/security

9. Research Paper:
   Lattice-Based Cryptography for Post-Quantum Security,
   IEEE Xplore Digital Library.

10. Research Paper:
    A Survey on Post-Quantum Cryptography and Its Applications,
    International Journal of Computer Applications (IJCA).